Public Co-Creation Hubs, Cross-Regional Pipelines, and Dynamic Efficiency in Thin Regional Ecosystems

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Öffentliche Co-Creation-Hubs, überregionale Pipelines und dynamische Effizienz in dünnen regionalen Ökosystemen

Abstract

Local governments increasingly support co-creation hubs as place-based instruments for strengthening regional innovation. Yet the welfare case for hub policy is unclear in thin regional ecosystems, where the local participant pool is limited and repeated local interaction may quickly become redundant. This paper develops a two-region, two-period framework in which a hub raises local matching in a target region, while pipeline policy creates cross-regional renewal through outside linkages. The model focuses on a wedge that arises when current hub services and directly brokered outside links are contractible, but the future novelty created inside the regional system is not. The framework delivers three conditional implications. First, decentralized support and planner evaluation need not coincide because renewal effort is underprovided and hub support can be either too weak or too strong. Second, a hub is socially efficient as a stand-alone intervention only under limited conditions. Third, pipeline policy has positive stand-alone value, but a hub-plus-pipeline bundle dominates pipeline policy alone only under specific parameter conditions. The framework generates empirically testable predictions on repeated ties, outside collaboration, and absorptive conditions, and it implies that hubs in thin regional ecosystems should be evaluated not only against no-policy baselines, but also against pipeline-only renewal strategies.

Zusammenfassung

Lokale Regierungen fördern zunehmend Co-Creation-Hubs als ortsbezogene Instrumente zur Stärkung regionaler Innovation. Ob Hub-Politik in dünnen regionalen Ökosystemen wohlfahrts-

steigernd ist, bleibt jedoch unklar, weil der lokale Teilnehmerpool begrenzt ist und sich wiederholende lokale Interaktionen rasch redundant werden können. Dieser Beitrag entwickelt ein Zwei-Regionen-Zwei-Perioden-Modell, in dem ein Hub das lokale Matching in einer Zielregion verbessert, während eine Pipeline-Politik durch externe Verbindungen überregionale Erneuerung schafft. Im Mittelpunkt steht ein Keil, der entsteht, wenn laufende Hub-Leistungen und direkt vermittelte externe Verbindungen vertraglich erfassbar sind, die künftig im regionalen System entstehende Neuheit jedoch nicht. Das Modell liefert drei bedingte Ergebnisse. Erstens können dezentrale Unterstützung und planerische Bewertung auseinanderfallen, weil Erneuerungsanstrengungen unterbereitgestellt werden und Hub-Unterstützung sowohl zu schwach als auch zu stark ausfallen kann. Zweitens ist ein Hub als isolierte Maßnahme nur unter begrenzten Bedingungen sozial effizient. Drittens hat Pipeline-Politik auch für sich genommen positiven Wert, doch ein Bündel aus Hub und Pipeline dominiert Pipeline-Politik allein nur in bestimmten Parameterbereichen. Daraus ergeben sich empirisch prüfbare Vorhersagen zu wiederholten Beziehungen, externer Zusammenarbeit und absorptiven Voraussetzungen. Für dünne regionale Ökosysteme folgt daraus, dass Hubs nicht nur gegen eine Situation ohne Politik, sondern auch gegen reine Pipeline-Strategien bewertet werden sollten.

Keywords: public co-creation hubs; cross-regional pipelines; regional innovation; place-based policy; thin regional ecosystems; dynamic efficiency

JEL classification: H70, O31, O38, R11, R12, R58

1 Introduction

Publicly supported co-creation hubs, enterprise hubs, and related collaborative spaces have become increasingly visible instruments of regional development policy. Local governments often view such hubs as a way to reduce interaction frictions among firms, researchers, entrepreneurs, and other regional actors by creating spaces and programs that facilitate meetings, information exchange, and collaboration. This policy logic is especially attractive in thin regional ecosystems, where the local pool of potentially complementary actors is limited and decentralized coordination may otherwise be weak. Yet the welfare case for hub policy is not obvious in such places. A hub may generate visible activity while still failing to create enough durable novelty to justify its cost, especially if the same local actors repeatedly interact within a narrow regional pool.

This tension sits naturally within the broader literature on agglomeration and regional coordination. A long tradition in urban and regional economics emphasizes that spatial concentration

can raise productivity through matching, learning, and related external economies (Duranton and Puga, 2004; Rosenthal and Strange, 2004). Face-to-face interaction has often been treated as a particularly important channel through which localized economic activity generates value (Storper and Venables, 2004). Related work on local buzz and global pipelines similarly suggests that geographically proximate interaction can be productive, especially when local contact is complemented by connections to outside knowledge and opportunities (Bathelt et al., 2004). More recent evidence from co-working environments also indicates that proximate social interaction can indeed create knowledge spillovers and collaboration gains (Roche et al., 2024). Taken together, these contributions provide a prima facie rationale for why a public hub might help a region.

At the same time, the same literature also implies that more local interaction is not always better. Proximity can support learning, but it can also generate lock-in when interaction becomes repetitive and the same actors repeatedly draw on the same local knowledge base (Boschma, 2005; Torre and Rallet, 2005). The regional innovation systems literature likewise stresses that local development depends not only on whether interaction occurs, but also on whether regional systems are able to renew themselves through access to external ideas, organizations, and opportunities (Cooke et al., 1997; Asheim et al., 2011; Tödtling and Trippl, 2005). This concern is especially relevant for place-based policy. Local interventions may correct coordination problems, but their effects are conditional and context-dependent rather than automatic (Kline and Moretti, 2014a; Neumark and Simpson, 2015). In the case of public hubs, this raises a simple but underexplored question: when should a hub be evaluated as a stand-alone project, and when should it instead be evaluated against an active renewal strategy that already builds outside links?

This paper develops a stylized two-region, two-period framework for public co-creation hubs in thin regional ecosystems. Region A is the target thin region in which a public hub may raise the local matching rate among potential collaborators. Region B is an outside pool from which additional collaborators, ideas, and partnerships may arrive. It should not be interpreted as a fully modeled second regional economy. Rather, it is a reduced-form outside access margin through which cross-regional renewal can occur. The model distinguishes two policy margins. The first is *local coordination*: a hub raises the rate at which potentially valuable local interactions in region A are realized. The second is *cross-regional renewal*: pipeline effort creates additional links between region A and outside actors in region B.

A key modeling choice is to distinguish decentralized local support from social evaluation. The paper focuses on a *dynamic coordination externality* that arises when current hub services and directly brokered outside links are contractible, but the future novelty created inside the regional system is not. Repeated local matching may exhaust novelty, while outside renewal can partially replenish it. The resulting wedge does not rely on an ad hoc short-run rule, but on a stylized

appropriability problem in which future local novelty behaves like a diffuse regional asset. The setup is dynamic only in a minimal two-period sense: it isolates one intertemporal wedge rather than modeling cumulative lock-in in full generality.

The framework delivers three conditional implications. First, decentralized support and planner evaluation need not coincide. Given the maintained appropriability structure, decentralized actors underprovide renewal effort because they do not fully internalize its contribution to future local novelty, and hub support can be either too weak or too strong depending on whether renewal gains or redundancy losses dominate. Second, a hub is socially efficient as a stand-alone intervention only under limited conditions: its local coordination gain must be large enough to offset fixed cost and the dynamic redundancy generated by repeated local interaction. Third, pipeline policy has positive stand-alone value, but the relevant policy object is often a *bundle* combining local coordination with cross-regional renewal. A bundle can dominate pipeline policy alone, but only under specific parameter conditions.

These results speak directly to current regional policy practice. In many settings, hubs are assessed using visible indicators such as participation counts, meetings, events, occupancy rates, or early project volume. The model suggests that such measures are informative about immediate coordination gains, but incomplete as welfare metrics. What matters for regional efficiency is not only whether a hub increases local interaction, but also whether novelty persists over time, whether outside renewal is adequately supplied, and whether the hub creates enough value relative to a pipeline-only alternative.

The paper's contribution is narrower than a general theory of regional development, but more pointed in policy terms. First, it redefines the relevant policy baseline: hub policy should often be evaluated not only against no intervention, but also against pipeline-only renewal. Second, it provides a minimal model that separates *local coordination* from *cross-regional renewal* as distinct policy margins. Third, it shows why visible local activity is an incomplete welfare metric when current coordination changes future novelty. In that sense, the paper speaks to the agglomeration, renewal, and place-based policy literatures by isolating a specific policy comparison they do not solve directly (Duranton and Puga, 2004; Bathelt et al., 2004; Boschma, 2005; Kline and Moretti, 2014a; Neumark and Simpson, 2015).

The model also generates empirically testable implications. It predicts that hub creation should raise initial interaction volume, but not necessarily preserve the share of new ties; that pipeline-only regions should exhibit more outside collaboration than no-policy regions; and that bundled regions should outperform pipeline-only regions when the outside pool is richer or regional absorptive capacity is stronger. These predictions point toward a dynamic and network-aware empirical evaluation of hub policy, one that tracks repeated ties, outside linkages, and participant composition

rather than relying only on visible activity measures.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature and situates the paper's contribution. Section 3 presents the two-region model and the dynamic coordination externality. Section 4 characterizes the conditions under which a hub is socially efficient as a stand-alone policy. Section 5 compares decentralized support with planner evaluation. Section 6 analyzes pipeline-alone policy, the bundled policy design, and the conditions under which the bundle is preferred. Section 7 develops extensions on partial internalization and participant composition. Section 8 derives empirical predictions and discusses regional policy interpretation. Section 9 concludes.

2 Related Literature and Contribution

This paper is positioned at the intersection of three literatures that are central to regional science: work on local interaction and regional coordination, work on proximity, lock-in, and system renewal, and work on the welfare evaluation of place-based policy. The paper's aim is not to offer a general theory of regional development. It is to provide a tractable policy framework for a specific regional question: when a publicly supported hub in a thin regional ecosystem should be evaluated as a stand-alone coordination device, and when it should instead be evaluated together with a policy that builds cross-regional pipelines.

2.1 Local Interaction, Thin Regional Ecosystems, and Coordination Devices

A first relevant literature explains why spatial concentration and local interaction can raise productivity. In urban and regional economics, agglomeration economies are commonly rationalized through matching, learning, and sharing mechanisms (Duranton and Puga, 2004; Rosenthal and Strange, 2004). A hub policy fits naturally into this tradition. It can be interpreted as a public coordination device that reduces local interaction frictions and raises the rate at which potentially valuable local matches are realized within a region.

This logic is reinforced by work emphasizing the distinctive role of face-to-face contact, tacit knowledge exchange, and proximate social interaction (Storper and Venables, 2004; Jaffe et al., 1993; Roche et al., 2024). At the same time, recent work on co-working spaces, enterprise hubs, and related collaborative venues in rural and peripheral settings suggests that these devices are highly context-dependent (Vogl and Akhavan, 2022; Merrell et al., 2021, 2022; OECD, 2025). Thin and nonmetropolitan regions may benefit from organized interaction, but the gains are neither automatic nor uniform.

The present paper takes this literature as a starting point, but shifts the question. Rather than asking whether local interaction matters, it asks when a *publicly supported* device for organizing local interaction is socially justified in a thin regional ecosystem. In that sense, the paper is less about agglomeration in general than about the policy evaluation of a specific coordination instrument.

2.2 Proximity, Lock-In, Regional Innovation Systems, and External Renewal

A second relevant literature emphasizes that proximity has both benefits and risks. Boschma (2005) and Torre and Rallet (2005) argue that proximity can support learning and collaboration, but can also generate lock-in when interaction becomes overly repetitive. This concern is especially important for the present paper because its principal distortion is dynamic rather than static. The model formalizes the idea that dense current local interaction may reduce the novelty available to the regional system in the future.

The paper also speaks to the regional innovation systems literature, which stresses that regional performance depends not only on local scale, but also on institutional structure, organizational capacity, and system renewal (Cooke et al., 1997; Asheim et al., 2011; Tödtling and Trippl, 2005; Coenen et al., 2017). Work on path renewal and path creation similarly emphasizes that regions often need mechanisms that reconnect them to new knowledge combinations and outside opportunities (Chaminade et al., 2019). In regional science terms, the key issue is not simply whether local interaction is dense, but whether a region can combine local coordination with access to external sources of renewal.

Bathelt et al. (2004)'s distinction between local buzz and global pipelines is especially useful here. The paper adopts that conceptual separation, but translates it into a policy comparison. The hub is modeled as a local coordination device inside the target region, whereas pipeline policy is modeled as a reduced-form mechanism that creates cross-regional renewal through outside links. This is the paper's main bridge to the knowledge-network and inter-regional collaboration literatures: it treats external linkages not as background context, but as a distinct policy margin that can be compared directly with local coordination in welfare terms. Empirical work on collaborative knowledge production supports the relevance of this distinction. Hoekman et al. (2009) show that inter-regional knowledge collaboration in Europe is strongly shaped by geographic and institutional distance, while Neuländtner et al. (2025) find that the benefits of inter-city R&D collaboration depend on place-based conditions and local knowledge endowments. Yet these literatures typically do not provide a welfare comparison among *hub alone*, *pipeline alone*, and the *bundle*. That is the narrower gap this paper tries to fill.

The extension on participant composition also connects to work on absorptive capacity and anchor actors. Regions differ not only in headcount, but also in whether they contain actors that can transform contact into learning and collaboration (Cohen and Levinthal, 1990; Agrawal and Cockburn, 2003). That extension remains secondary to the baseline model, but it reinforces the regional-science point that the effective thickness of an ecosystem depends on composition as well as size.

2.3 Place-Based Policy, Welfare Evaluation, and This Paper's Contribution

A third relevant literature studies the welfare properties of geographically targeted policy. Kline and Moretti (2014a) provide a general welfare framework for local economic development policy, and Kline and Moretti (2014b) show in a major historical setting that place-based interventions can have durable local effects. More broadly, Neumark and Simpson (2015) stress that the case for place-based policy depends on the underlying distortion and on the relevant margins of adjustment. That perspective is directly relevant here. A public hub is a place-based intervention, but its value depends on more than visible local activity. It depends on whether the intervention corrects a meaningful regional distortion and on how it interacts with external renewal.

The paper also speaks to a more specific literature on collaborative innovation policy. Here the relevant question is not only whether public support raises innovation, but whether policy-induced collaboration helps regions access external knowledge and diversify into new activities. The evidence is informative but not uniformly optimistic. Bednarz and Broekel (2019) show that policy-induced R&D networks need not automatically translate into stronger inter-regional knowledge diffusion, while Mewes and Broekel (2020) find that subsidized joint R&D projects can be more effective for regional technological diversification than subsidies to single organizations. For the present paper, the important point is that collaborative policy should be evaluated relative to the specific regional margin it changes. Existing work tells us that collaborative support can matter, but it does not solve the baseline problem studied here: whether a publicly supported hub should be judged against no policy or against a renewal policy that already builds outside links.

The paper contributes to this literature in three specific ways. First, it changes the relevant policy baseline: because pipeline policy has positive stand-alone value, the appropriate benchmark is often not hub versus no hub, but pipeline alone versus a bundle that combines local coordination with external renewal. Second, it introduces a clean regional distinction between *local coordination* and *cross-regional renewal* as separate policy margins. Third, it derives a welfare warning from a single principal wedge, a *dynamic coordination externality*: visible local activity need not be an adequate metric when present interaction changes the future novelty of the regional system. The

paper is therefore less a general theory of regional development than a stylized policy comparison targeted at one specific gap in the literature.

This contribution is deliberately stylized. The model does not attempt to explain endogenous network formation, migration, or location choice in full generality. Instead, it provides a tractable regional policy framework that isolates a single welfare wedge and shows how that wedge changes the evaluation of hub policy in thin regional ecosystems. That is the sense in which the paper is intended as a policy-relevant stylized framework for regional science rather than as a broad structural theory of regional development.

3 The Model

We consider a target thin region A and an outside region B. A hub policy in region A works on a *local coordination* margin by raising the rate at which potentially valuable local interactions are realized. A pipeline policy works on a *cross-regional renewal* margin by creating additional links between region A and outside actors in region B. The model is dynamic only in a minimal two-period sense: it is designed to isolate one intertemporal welfare wedge, not to provide a full theory of cumulative regional path dependence.

The section also introduces the paper's principal distortion. The key wedge is a *dynamic coordination externality*: decentralized actors value the immediate local matches generated by stronger coordination, but they do not fully internalize how dense current local interaction changes the common stock of future novelty available to the regional system. Repeated local interaction may exhaust novelty, while cross-regional renewal partially replenishes it.

3.1 Environment and Policy Instruments

Region A contains $N_A \geq 2$ local participants, where a participant may be interpreted as a firm, researcher, entrepreneur, or other actor capable of generating collaborative surplus. Region B contains $N_B \geq 1$ outside participants who form the external pool from which new ideas, collaborators, and partnerships may arrive. Region B is not modeled as a symmetric second local economy. It is a reduced-form outside access margin through which cross-regional renewal can occur.

The number of unordered potential *local* pairs within region A is

$$P_A := \frac{N_A(N_A - 1)}{2}, \tag{1}$$

and the number of potential *cross-regional* links between regions A and B is

$$Q_{AB} := N_A N_B. \tag{2}$$

Two policy instruments are available. First, region A may establish a public co-creation hub,

$$h \in \{0, 1\},$$

where $h = 1$ means that the hub is established. Second, the policymaker or decentralized local coalition may choose cross-regional pipeline effort

$$b \geq 0.$$

The variable b summarizes brokerage, interregional outreach, visiting programs, and other efforts that connect region A to outside actors.

Time is discrete and consists of two periods, $t = 1, 2$. The hub affects local matching in both periods. Pipeline effort affects only period 2.

3.2 Local Coordination and Period-1 Surplus

Let $m_A(h)$ denote the effective local matching rate among potential pairs within region A. We assume

$$m_A(h) = m_A^0 + \left(m_A^1 - m_A^0\right)h, \qquad 0 \leq m_A^0 < m_A^1 \leq 1. \tag{3}$$

Thus, establishing a hub lowers local search and meeting frictions in region A and raises the fraction of potential local pairs that become realized economically meaningful interactions.

Let $v_L > 0$ denote the average social surplus generated by one realized and still-novel local interaction. Period-1 gross surplus is therefore

$$B_1(h) = v_L P_A m_A(h). \tag{4}$$

3.3 Dynamic Coordination Externality

The principal distortion in the model arises because the same local interactions that create value in period 1 can reduce the novelty available to the regional system in period 2. To capture this

parsimoniously, let the period-2 *local* component of social surplus be

$$B_{2L}(h,b) = \beta v_L P_A m_A(h)\big[1 - \delta m_A(h) + \xi b\big], \qquad \beta \in (0,1],\ \delta > 0,\ \xi > 0. \tag{5}$$

The term $\delta m_A(h)$ captures dynamic redundancy: more intensive local matching in period 1 raises the chance that period-2 interaction becomes repetitive and less novel. The term ξb captures the idea that cross-regional renewal partially replenishes the local opportunity set by bringing in outside ideas, collaborators, or perspectives.

We maintain the regularity condition

$$1 - \delta m_A^1 > 0, \tag{6}$$

which ensures that second-period local novelty remains positive even without pipeline effort.

3.4 Cross-Regional Links and a Minimally Active Outside Region

Let the effective cross-regional connection intensity be

$$q(h,b) = q_0 + \big(\eta_0 + \eta_1 h\big)b, \qquad q_0 \geq 0, \quad \eta_0 > 0, \quad \eta_1 \geq 0. \tag{7}$$

The baseline term q_0 captures outside access that exists even without new policy. The parameter $\eta_0 > 0$ implies that pipeline effort can create value even without a hub: region B is therefore not a purely passive outside pool. The complementarity term $\eta_1 h$ captures the idea that a hub in region A can make brokerage more productive by organizing, absorbing, and channeling outside contacts.

Let $v_X > 0$ denote the average surplus generated by one effective cross-regional linkage. The period-2 *external* component of surplus is then

$$B_{2X}(h,b) = \beta v_X Q_{AB} q(h,b). \tag{8}$$

Combining (5) and (8), total period-2 gross surplus is

$$B_2(h,b) = \beta\Big\{v_L P_A m_A(h)\big[1 - \delta m_A(h) + \xi b\big] + v_X Q_{AB}\big[q_0 + \big(\eta_0 + \eta_1 h\big)b\big]\Big\}. \tag{9}$$

3.5 Social Welfare, Contractibility, and the Decentralized Support Benchmark

Establishing the hub requires a fixed cost $F > 0$. Pipeline effort has convex cost

$$\frac{\kappa}{2}b^2, \qquad \kappa > 0.$$

Total social welfare is therefore

$$W(h, b) = B_1(h) + B_2(h, b) - Fh - \frac{\kappa}{2}b^2. \tag{10}$$

Using (4) and (9), this becomes

$$W(h, b) = v_L P_A m_A(h) + \beta\Big\{v_L P_A m_A(h)\big[1 - \delta m_A(h) + \xi b\big] + v_X Q_{AB}\big[q_0 + (\eta_0 + \eta_1 h)b\big]\Big\} - Fh - \frac{\kappa}{2}b^2. \tag{11}$$

To isolate the principal distortion, we introduce a decentralized support benchmark. Think of a local coalition, private operator, or decentralized set of sponsors that can contract on current hub services and on directly brokered outside links, but cannot fully contract on the continuation value of the regional opportunity set. The institutional logic is intentionally simple. Current meetings, memberships, and brokered introductions are observable and can therefore be priced. By contrast, once present interaction changes who knows whom inside the region, the resulting future novelty becomes a diffuse regional asset: later matches among third parties are hard to verify ex ante, exclusion is difficult, and much of the benefit takes the form of noncontractible option value. In that sense, the continuation value of future local novelty is treated as a local public-good or common-pool asset rather than as something owned by the current sponsoring coalition.

To keep the baseline sharp, the decentralized benchmark studies the polar case in which this continuation value is not internalized at all. Section 7 relaxes that benchmark through partial internalization. Under this polar benchmark, the decentralized objective is

$$\Pi(h, b) = v_L P_A m_A(h) + \beta v_X Q_{AB}\big[q_0 + (\eta_0 + \eta_1 h)b\big] - Fh - \frac{\kappa}{2}b^2. \tag{12}$$

The difference between (11) and (12) is exactly the common continuation value generated by the local regional system:

$$\beta v_L P_A m_A(h)\big[1 - \delta m_A(h) + \xi b\big].$$

This is the paper's dynamic coordination externality in the baseline benchmark. Stronger local coordination creates private current benefits, but its effects on the future novelty of the local

regional system are not fully priced by decentralized actors. The benchmark should therefore be read as a maintained appropriability structure, not as a claim that all decentralized settings literally involve zero internalization.

For later use, define

$$A_h^P := v_X Q_{AB}(\eta_0 + \eta_1 h), \tag{13}$$

and

$$A_h^W := \xi v_L P_A m_A(h) + A_h^P. \tag{14}$$

Then (12) and (11) can be written as

$$\Pi(h, b) = \bar{\Pi}(h) + \beta A_h^P b - \frac{\kappa}{2} b^2, \qquad \bar{\Pi}(h) := v_L P_A m_A(h) + \beta v_X Q_{AB} q_0 - Fh, \tag{15}$$

and

$$W(h, b) = \bar{W}(h) + \beta A_h^W b - \frac{\kappa}{2} b^2, \tag{16}$$

where

$$\bar{W}(h) := v_L P_A m_A(h) + \beta \Big\{ v_L P_A m_A(h) \big[1 - \delta m_A(h)\big] + v_X Q_{AB} q_0 \Big\} - Fh. \tag{17}$$

Since both (15) and (16) are strictly concave in b, the optimal pipeline choices conditional on h are well defined. Section 5 uses this benchmark to show why decentralized support and social evaluation need not coincide. Section 4 first isolates the stand-alone hub comparison by setting $b = 0$, and Section 6 then restores pipeline effort to compare pipeline-alone, hub-alone, and bundled policy designs.

4 When Is a Public Hub Efficient?

This section isolates when a public hub is socially efficient *as a stand-alone local policy*. Throughout, we set

$$b = 0.$$

This makes it possible to evaluate the hub's local coordination value before introducing pipeline policy in Section 6.

4.1 A Benchmark Efficiency Criterion

Using the period-1 and period-2 local surplus expressions from Section 3, the net welfare gain from establishing a hub without pipeline policy is

$$\Delta_H = v_L P_A \left(m_A^1 - m_A^0\right)\left[(1+\beta) - \beta\delta\left(m_A^1 + m_A^0\right)\right] - F. \tag{18}$$

It is useful to define the hub's *dynamic viability factor* as

$$\Lambda_A := (1+\beta) - \beta\delta\left(m_A^1 + m_A^0\right). \tag{19}$$

Then (18) becomes

$$\Delta_H = v_L P_A \left(m_A^1 - m_A^0\right)\Lambda_A - F. \tag{20}$$

The term $m_A^1 - m_A^0$ captures the hub's immediate coordination gain, while Λ_A summarizes whether that gain survives second-period redundancy. A larger δ lowers Λ_A and therefore weakens the case for a hub.

The hub is socially efficient if and only if

$$\Delta_H \geq 0.$$

Proposition 1. *Consider the benchmark case without pipeline policy, $b = 0$.*

1. *If*

$$\Lambda_A \leq 0,$$

then the hub is never socially efficient:

$$\Delta_H < 0 \qquad \textit{for all } N_A \geq 2 \textit{ and } F > 0.$$

2. *If*

$$\Lambda_A > 0,$$

then the hub is socially efficient if and only if

$$P_A \geq \frac{F}{v_L\left(m_A^1 - m_A^0\right)\Lambda_A}. \tag{21}$$

Equivalently, treating N_A as continuous, the hub is efficient if and only if

$$N_A \geq N_H^* := \frac{1}{2}\left(1 + \sqrt{1 + \frac{8F}{v_L\left(m_A^1 - m_A^0\right)\Lambda_A}}\right). \tag{22}$$

If N_A is restricted to integers, replace N_H^* by its ceiling.

Proposition 1 says that a hub creates value only if the gross local coordination benefit,

$$v_L P_A\left(m_A^1 - m_A^0\right),$$

survives the dynamic penalty summarized by Λ_A and is large enough to cover F. Failure can therefore come either from thin local scale or from strong dynamic redundancy.

An implication of the two-region setup is that the stand-alone criterion in (20) depends only on *local* primitives. Outside-pool variables affect welfare levels, but they cancel out of the incremental comparison between a hub alone and no hub.

4.2 Equivalent Threshold Characterizations

Proposition 1 can be rewritten in two equivalent threshold forms.

First, for given $(N_A, m_A^0, m_A^1, \beta, \delta)$, define the critical fixed cost

$$F_H^*(N_A) := v_L P_A\left(m_A^1 - m_A^0\right)\Lambda_A. \tag{23}$$

Then the hub is efficient if and only if

$$F \leq F_H^*(N_A).$$

Second, for given $(N_A, m_A^0, m_A^1, \beta, F)$, define the critical redundancy rate

$$\delta_H^*(N_A) := \frac{1+\beta}{\beta\left(m_A^1 + m_A^0\right)} - \frac{F}{\beta v_L P_A\left(m_A^1 - m_A^0\right)\left(m_A^1 + m_A^0\right)}. \tag{24}$$

Then the hub is efficient if and only if

$$\delta \leq \delta_H^*(N_A),$$

provided the right-hand side is positive.

These thresholds show that stand-alone hub efficiency is governed by five local primitives: N_A, $m_A^1 - m_A^0$, v_L, F, and δ.

4.3 Comparative Statics

The next proposition summarizes the main comparative statics.

Proposition 2. *Suppose $\Lambda_A > 0$. Then the hub-alone welfare gain Δ_H satisfies*

$$\frac{\partial \Delta_H}{\partial N_A} = \frac{v_L}{2}(2N_A - 1)\left(m_A^1 - m_A^0\right)\Lambda_A > 0, \tag{25}$$

$$\frac{\partial \Delta_H}{\partial F} = -1 < 0, \tag{26}$$

$$\frac{\partial \Delta_H}{\partial \delta} = -\beta v_L P_A \left(m_A^1 - m_A^0\right)\left(m_A^1 + m_A^0\right) < 0. \tag{27}$$

Moreover, under the regularity condition (6)*,*

$$\frac{\partial \Delta_H}{\partial m_A^1} = v_L P_A \left[(1+\beta) - 2\beta\delta m_A^1\right] > 0, \tag{28}$$

$$\frac{\partial \Delta_H}{\partial m_A^0} = -v_L P_A \left[(1+\beta) - 2\beta\delta m_A^0\right] < 0. \tag{29}$$

Proposition 2 yields three immediate implications. First, the hub is more likely to be efficient in a thicker target ecosystem. Since

$$P_A = \frac{N_A(N_A - 1)}{2},$$

the number of potential local pairs rises more than proportionally with N_A. Thin ecosystems are therefore disadvantaged even when the hub raises matching.

Second, faster local redundancy makes the hub less attractive. This is the model's reduced-form representation of dynamic redundancy risk from repeated interaction inside region A.

Third, a larger matching effect m_A^1 raises the hub's value, while a higher baseline m_A^0 lowers its incremental contribution. The hub is therefore most useful where pre-existing coordination is weak but the policy can materially reduce frictions.

4.4 Discussion

Two points matter for what follows. Hub efficiency is not determined by local scale alone, because strong redundancy can overturn even sizable short-run coordination gains. And the relevant object

is Λ_A, not just the static increment $m_A^1 - m_A^0$. A hub may therefore look successful on visible activity grounds while remaining dynamically weak. The next section compares that stand-alone logic with decentralized support under the model's dynamic coordination externality.

5 Why Public Intervention? Decentralized Support and Social Evaluation

Section 4 characterized when a hub is socially efficient as a stand-alone policy. The next question is why decentralized local support may diverge from that social evaluation. In this framework, the divergence comes from the contractibility structure in Section 3: decentralized actors do not fully internalize how current local coordination changes the common stock of future novelty available to the regional system.

5.1 Decentralized Renewal Choice Versus the Planner

From (15), a decentralized local coalition choosing pipeline effort conditional on hub state h solves

$$\max_{b\geq 0} \Pi(h,b) = \max_{b\geq 0} \left\{ \bar{\Pi}(h) + \beta A_h^P b - \frac{\kappa}{2} b^2 \right\}.$$

From (16), the planner solves

$$\max_{b\geq 0} W(h,b) = \max_{b\geq 0} \left\{ \bar{W}(h) + \beta A_h^W b - \frac{\kappa}{2} b^2 \right\}.$$

Proposition 3. *For each $h \in \{0,1\}$, the decentralized support benchmark and the planner choose*

$$b_P^*(h) = \frac{\beta A_h^P}{\kappa} = \frac{\beta v_X Q_{AB}\big(\eta_0 + \eta_1 h\big)}{\kappa}, \tag{30}$$

and

$$b_W^*(h) = \frac{\beta A_h^W}{\kappa} = \frac{\beta\Big[\xi v_L P_A m_A(h) + v_X Q_{AB}\big(\eta_0 + \eta_1 h\big)\Big]}{\kappa}, \tag{31}$$

respectively. Hence

$$b_W^*(h) - b_P^*(h) = \frac{\beta \xi v_L P_A m_A(h)}{\kappa} > 0. \tag{32}$$

Moreover,

$$\big[b_W^*(1) - b_P^*(1)\big] - \big[b_W^*(0) - b_P^*(0)\big] = \frac{\beta \xi v_L P_A \big(m_A^1 - m_A^0\big)}{\kappa} > 0. \tag{33}$$

Proposition 3 is a direct implication of the maintained appropriability structure. Once decentralized actors value only A_h^P while the planner values $A_h^W = A_h^P + \xi v_L P_A m_A(h)$, the planner necessarily chooses more renewal. The gap is larger when a hub exists because denser local interaction raises the social value of replenishing the regional opportunity set.

5.2 Hub Support Under Decentralized and Social Evaluation

Substituting (30) and (31) back into the two objectives yields

$$\Pi^*(h) := \Pi\big(h, b_P^*(h)\big) = \bar{\Pi}(h) + R_h^P, \qquad R_h^P := \frac{\beta^2 (A_h^P)^2}{2\kappa}, \tag{34}$$

and

$$W^*(h) := W\big(h, b_W^*(h)\big) = \bar{W}(h) + R_h^W, \qquad R_h^W := \frac{\beta^2 (A_h^W)^2}{2\kappa}. \tag{35}$$

Define the incremental hub-support criteria

$$\Delta_P^* := \Pi^*(1) - \Pi^*(0), \tag{36}$$

and

$$\Delta_W^* := W^*(1) - W^*(0). \tag{37}$$

For notational convenience, let

$$S_A := v_L P_A \big(m_A^1 - m_A^0\big) > 0, \tag{38}$$

denote the hub's immediate local coordination gain, and let

$$D_A := \beta v_L P_A \big(m_A^1 - m_A^0\big)\Big[1 - \delta\big(m_A^1 + m_A^0\big)\Big] \tag{39}$$

denote the dynamic local correction from Section 4. Finally, define the incremental renewal wedge

$$M_A := \big(R_1^W - R_0^W\big) - \big(R_1^P - R_0^P\big) > 0. \tag{40}$$

Using (34)–(40), the decentralized and social support criteria can be written as

$$\Delta_P^* = S_A - F + \big(R_1^P - R_0^P\big), \tag{41}$$

and

$$\Delta_W^* = \Delta_P^* + D_A + M_A. \tag{42}$$

Equation (42) decomposes the planner–decentralized wedge into two parts: D_A, the dynamic value or cost of denser local coordination, and M_A, the added social value created when the planner chooses more renewal than decentralized actors do. This is an accounting decomposition under the benchmark contract structure from Section 3.

Proposition 4. *Define*

$$F_P^* := S_A + \left(R_1^P - R_0^P\right), \tag{43}$$

and

$$F_W^* := F_P^* + D_A + M_A. \tag{44}$$

1. *If*

$$D_A + M_A > 0, \tag{45}$$

then there exists a nonempty fixed-cost interval

$$F_P^* < F < F_W^* \tag{46}$$

such that decentralized actors reject the hub while the planner supports it.

2. *If*

$$D_A + M_A < 0, \tag{47}$$

then there exists a nonempty fixed-cost interval

$$F_W^* < F < F_P^* \tag{48}$$

such that decentralized actors support the hub while the planner rejects it.

Proposition 4 gives the model's minimal policy implication under the maintained appropriability structure. If $D_A + M_A > 0$, decentralized actors under-support the hub because they neglect part of the continuation value created by renewal. If $D_A + M_A < 0$, they over-support it because they see current coordination gains without fully pricing the future novelty loss from repeated local interaction. The direction of policy error is therefore parameter-dependent rather than fixed in advance.

5.3 Interpretation

Three implications follow. First, the decentralized benchmark is not an ad hoc activity rule: it comes from an appropriability problem in which current coordination and directly brokered outside links are contractible, but the diffuse continuation value of future novelty is not. Section 7 shows that this wedge shrinks under partial internalization and that the associated support-threshold gap converges to zero as internalization approaches one. Second, the principal distortion is dynamic rather than static: decentralized actors do not misread period-1 coordination, but underprice what it does to the future opportunity set. Third, the distortion links local coordination and cross-regional renewal. Because the planner values how outside links replenish novelty, renewal is underprovided precisely where hub-supported coordination is strongest.

6 Pipeline Alone, Hub Alone, and the Efficient Policy Bundle

Sections 4 and 5 established two points: a hub alone may fail because stronger local coordination also raises dynamic redundancy risk, and decentralized actors underprovide renewal because they do not fully internalize its contribution to future novelty. This section therefore compares the relevant policy designs directly: *pipeline alone*, *hub alone*, and the *bundle*.

6.1 Optimal Pipeline Policy Conditional on Hub State

From (16), conditional on hub state h, the planner chooses

$$\max_{b\geq 0} W(h,b) = \max_{b\geq 0}\left\{\bar{W}(h) + \beta A_h^W b - \frac{\kappa}{2}b^2\right\}.$$

Proposition 5. *For each $h \in \{0,1\}$, the planner's unique welfare-maximizing pipeline policy is*

$$b^*(h) = b_W^*(h) = \frac{\beta A_h^W}{\kappa} = \frac{\beta\Big[\xi v_L P_A m_A(h) + v_X Q_{AB}\big(\eta_0 + \eta_1 h\big)\Big]}{\kappa}. \tag{49}$$

The associated maximized welfare level is

$$W^*(h) := W\big(h, b^*(h)\big) = \bar{W}(h) + R_h, \qquad R_h := \frac{\beta^2 (A_h^W)^2}{2\kappa}. \tag{50}$$

The term R_h is the maximized renewal surplus conditional on hub state h. Because

$$A_1^W - A_0^W = \xi v_L P_A \left(m_A^1 - m_A^0\right) + v_X Q_{AB} \eta_1 > 0,$$

we have

$$R_1 > R_0. \tag{51}$$

Thus, pipeline effort is valuable without a hub and more valuable when a hub is present. This inequality is a direct implication of the maintained benchmark and complementarity assumptions. The substantive question is whether the incremental renewal margin is large enough to offset the hub's stand-alone defect.

6.2 Pipeline Alone Versus the Bundle

If no policy is used, welfare is

$$W(0,0) = v_L P_A m_A^0 + \beta \left\{ v_L P_A m_A^0 \left(1 - \delta m_A^0\right) + v_X Q_{AB} q_0 \right\}. \tag{52}$$

If the planner uses pipeline policy without creating a hub, welfare is

$$W^*(0) = W(0,0) + R_0. \tag{53}$$

If the planner creates a hub but sets $b = 0$, welfare is

$$W(1,0) = W(0,0) + \Delta_H, \tag{54}$$

where Δ_H is the hub-alone gain from Section 4.

If the planner chooses the full bundle, welfare is

$$W^*(1) = W(1,0) + R_1 = W(0,0) + \Delta_H + R_1. \tag{55}$$

Proposition 6. *Assume $A_0^W > 0$. Then:*

1. ***Pipeline-alone value.*** *Pipeline policy without a hub is socially valuable relative to no intervention:*

$$W^*(0) - W(0,0) = R_0 > 0. \tag{56}$$

2. ***Bundle versus pipeline alone.*** *The hub-plus-pipeline bundle is socially preferred to pipeline*

policy alone if and only if

$$\Delta_H + \big(R_1 - R_0\big) \geq 0. \tag{57}$$

Equivalently, using $F_H^*(N_A)$ from Section 4, the bundle is preferred if and only if

$$F \leq F_{H+b}^{\dagger}(N_A, N_B) := F_H^*(N_A) + \big(R_1 - R_0\big). \tag{58}$$

3. ***Threshold expansion relative to hub alone.*** *Since $R_1 - R_0 > 0$, the bundle enlarges the parameter region in which hub creation is justified:*

$$F_{H+b}^{\dagger}(N_A, N_B) - F_H^*(N_A) = R_1 - R_0 > 0. \tag{59}$$

Proposition 6 contains the paper's central bundle comparison. Pipeline policy is already valuable without a hub, so the key condition is (57): the bundle is justified only when the hub's stand-alone contribution is not too negative relative to the incremental renewal margin $R_1 - R_0$.

Part (3) is best read as an algebraic restatement of that condition in threshold form. It is useful for visualization, but the substantive comparison is between *pipeline alone* and the *bundle*.

Proposition 7. *Suppose $R_1 > R_0$. Then there exists a positive fixed-cost interval for which a hub alone is inefficient but the hub-plus-pipeline bundle is preferred to pipeline policy alone. The interval is*

$$F_H^*(N_A) < F \leq F_H^*(N_A) + \big(R_1 - R_0\big). \tag{60}$$

Proposition 7 sharpens the policy comparison. Once pipeline effort has stand-alone value, the relevant planner choice is often not *hub* versus *no hub*, but *pipeline alone* versus a *hub embedded in a renewal strategy*. The bundle can rescue the hub only when the incremental coordination value of adding the hub is large enough. The result is therefore conditional and partly structure-driven: the benchmark assumptions generate an incremental renewal term, and the question is whether it is quantitatively strong enough to overcome the hub's stand-alone weakness.

6.3 Comparative Interpretation

Three comparative-static lessons matter most. The bundle becomes more attractive when outside renewal is more productive, for example because the outside pool is larger, brokerage is more effective, or outside links generate more surplus. It also becomes more attractive when renewal is more tightly connected to local novelty, that is, when ξ is larger. By contrast, a higher κ makes the bundle less attractive by lowering both R_0 and R_1 and compressing the difference between them.

6.4 Discussion

The central lesson is that cross-regional pipelines change the relevant baseline for evaluating a hub. Once pipeline policy has positive stand-alone value, the key question is whether adding the hub creates enough extra coordination benefit to justify its fixed cost and dynamic redundancy risk. In thin regional ecosystems, a hub intensifies local matching but may also intensify repetition, while pipeline policy works on the renewal margin. The efficient policy object is therefore often a bundle, but not automatically so. The operative condition is (57); (58) is simply its threshold representation. Section 7 shows that this bundle-support gap narrows continuously when decentralized actors internalize more of the continuation value.

7 Extensions

This section develops two extensions that sharpen the interpretation of the baseline model while preserving analytical tractability. The first relaxes the baseline polar assumption that decentralized actors internalize none of the continuation value of future local novelty. The second introduces heterogeneous absorptive capacity among participants in the target region. The aim is not to multiply cases, but to show how the paper's main policy comparison changes when the wedge is weakened and when ecosystem composition varies.

To streamline notation, define

$$\Delta m_A := m_A^1 - m_A^0 > 0, \qquad \Lambda_A := (1+\beta) - \beta\delta(m_A^1 + m_A^0),$$

so that in the baseline model

$$\Delta_H = v_L P_A \Delta m_A \Lambda_A - F, \qquad b_W^*(1) = \frac{\beta\Big[\xi v_L P_A m_A^1 + v_X Q_{AB}(\eta_0 + \eta_1)\Big]}{\kappa}.$$

We also retain the notation S_A and D_A from Section 5.

7.1 Partial Internalization of Future Novelty

The baseline benchmark in Section 3 takes the polar case in which decentralized actors internalize none of the continuation value generated inside the regional system. A natural robustness check is to allow partial internalization.

Let $\alpha \in [0, 1]$ denote the fraction of period-2 local continuation value that decentralized actors

internalize. Their objective becomes

$$\Pi_\alpha(h,b) = v_L P_A m_A(h) + \alpha\beta v_L P_A m_A(h)\big[1-\delta m_A(h)+\xi b\big] + \beta v_X Q_{AB}\big[q_0+(\eta_0+\eta_1 h)b\big] - Fh - \frac{\kappa}{2}b^2. \tag{61}$$

The baseline model corresponds to $\alpha = 0$, while $\alpha = 1$ is the full-internalization limit.

Define

$$A^P_{h,\alpha} := \alpha\xi v_L P_A m_A(h) + v_X Q_{AB}\big(\eta_0 + \eta_1 h\big), \tag{62}$$

so that (61) can be written as

$$\Pi_\alpha(h,b) = \bar{\Pi}_\alpha(h) + \beta A^P_{h,\alpha} b - \frac{\kappa}{2}b^2, \tag{63}$$

where

$$\bar{\Pi}_\alpha(h) := v_L P_A m_A(h) + \alpha\beta v_L P_A m_A(h)\big[1-\delta m_A(h)\big] + \beta v_X Q_{AB} q_0 - Fh. \tag{64}$$

Proposition 8. *For each $h \in \{0,1\}$ and each $\alpha \in [0,1]$, the decentralized benchmark with partial internalization chooses*

$$b^*_{P,\alpha}(h) = \frac{\beta A^P_{h,\alpha}}{\kappa} = \frac{\beta\Big[\alpha\xi v_L P_A m_A(h) + v_X Q_{AB}\big(\eta_0+\eta_1 h\big)\Big]}{\kappa}. \tag{65}$$

Hence

$$b^*_W(h) - b^*_{P,\alpha}(h) = \frac{\beta(1-\alpha)\xi v_L P_A m_A(h)}{\kappa} \geq 0, \tag{66}$$

with strict inequality for every $\alpha < 1$. Moreover,

$$\frac{\partial b^*_{P,\alpha}(h)}{\partial\alpha} = \frac{\beta\xi v_L P_A m_A(h)}{\kappa} > 0. \tag{67}$$

Proposition 8 shows that the baseline wedge is not an all-or-nothing knife edge. As decentralized actors internalize more of the continuation value, they choose more renewal and the planner–decentralized gap shrinks continuously. But for every $\alpha < 1$, renewal remains underprovided. The policy comparison in the main text therefore survives when the polar benchmark is weakened.

To connect this extension directly to the main policy comparison, define

$$R^P_{h,\alpha} := \frac{\beta^2 (A^P_{h,\alpha})^2}{2\kappa}, \qquad h \in \{0,1\}, \tag{68}$$

and let

$$F^\dagger_{P,\alpha} := S_A + \alpha D_A + \big(R^P_{1,\alpha} - R^P_{0,\alpha}\big) \tag{69}$$

denote the decentralized support threshold under partial internalization. The planner's corresponding threshold remains

$$F^{\dagger}_{H+b} = S_A + D_A + \big(R_1 - R_0\big), \tag{70}$$

which is the bundle threshold from Section 6.

Proposition 9. *Define*

$$X_A := \xi^2 v_L^2 P_A^2 \big((m_A^1)^2 - (m_A^0)^2\big) > 0 \tag{71}$$

and

$$Y_A := 2\xi v_L P_A v_X Q_{AB} \Big[\eta_0 \big(m_A^1 - m_A^0\big) + \eta_1 m_A^1\Big] > 0. \tag{72}$$

Then the planner–decentralized threshold gap under partial internalization is

$$\Omega_\alpha := F^{\dagger}_{H+b} - F^{\dagger}_{P,\alpha} = (1-\alpha)\left[D_A + \frac{\beta^2}{2\kappa}\Big((1+\alpha)X_A + Y_A\Big)\right]. \tag{73}$$

Hence:

1. *If $\Omega_\alpha > 0$, there exists a positive fixed-cost interval*

$$F^{\dagger}_{P,\alpha} < F \le F^{\dagger}_{H+b} \tag{74}$$

   in which decentralized actors reject hub creation while the planner prefers the bundle.

2. *If $\Omega_\alpha < 0$, there exists a positive fixed-cost interval*

$$F^{\dagger}_{H+b} < F \le F^{\dagger}_{P,\alpha} \tag{75}$$

   in which decentralized actors support hub creation while the planner rejects it.

3. *The threshold gap collapses in the full-internalization limit:*

$$\Omega_1 = 0. \tag{76}$$

   Moreover, if $D_A \ge 0$, then Ω_α is strictly decreasing in α on $[0, 1)$.

Proposition 9 strengthens the role of the extension. Partial internalization does not merely shrink the renewal-choice gap in (66); it also narrows the support-threshold gap that underlies Proposition 4 in Section 5 and the bundle comparison in Section 6. In the full-internalization limit, the decentralized and planner thresholds coincide. Under the empirically relevant case $D_A \ge 0$, that convergence is monotone.

7.2 Regional Absorptive Capacity and Anchor Actors

The baseline model treats all local participants symmetrically through the common surplus parameter v_L and the common cross-regional surplus parameter v_X. A natural extension allows some local actors in region A to have higher absorptive capacity or stronger boundary-spanning ability. This captures the idea that a region with the same number of participants may nevertheless generate more value if some actors are especially effective at transforming contact into productive collaboration.

Let $\sigma \in [0, 1]$ denote the share of *high-absorptive* participants in region A. Suppose that a realized local interaction yields surplus v_L^H if at least one of the two participants is high-absorptive and v_L^L otherwise, where

$$v_L^H > v_L^L > 0.$$

Then the expected surplus per realized local interaction is

$$\bar{v}_L(\sigma) = v_L^L(1-\sigma)^2 + v_L^H\left[1-(1-\sigma)^2\right] = v_L^L + (2\sigma - \sigma^2)(v_L^H - v_L^L). \tag{77}$$

Because

$$\bar{v}_L'(\sigma) = 2(1-\sigma)(v_L^H - v_L^L) > 0 \qquad \text{for } \sigma \in [0, 1), \tag{78}$$

a higher share of high-absorptive participants raises the average value of a realized local interaction.

Likewise, suppose that the social surplus from an effective cross-regional link is v_X^H when the region-A participant involved is high-absorptive and v_X^L otherwise, where

$$v_X^H > v_X^L > 0.$$

Then the expected surplus per effective cross-regional link is

$$\bar{v}_X(\sigma) = v_X^L(1-\sigma) + v_X^H\sigma = v_X^L + \sigma(v_X^H - v_X^L), \tag{79}$$

with

$$\bar{v}_X'(\sigma) = v_X^H - v_X^L > 0. \tag{80}$$

Replacing v_L by $\bar{v}_L(\sigma)$ and v_X by $\bar{v}_X(\sigma)$, the hub-alone welfare gain becomes

$$\Delta_H(\sigma) = \bar{v}_L(\sigma)P_A\Delta m_A\Lambda_A - F, \tag{81}$$

and the optimal pipeline policy conditional on a hub becomes

$$b^*(\sigma) = \frac{\beta A_1(\sigma)}{\kappa}, \qquad A_1(\sigma) := \xi\bar{v}_L(\sigma)P_A m_A^1 + \bar{v}_X(\sigma)Q_{AB}(\eta_0 + \eta_1), \tag{82}$$

with corresponding renewal surplus

$$R_1(\sigma) = \frac{\beta^2 A_1(\sigma)^2}{2\kappa}. \tag{83}$$

Proposition 10. *Suppose $v_L^H > v_L^L > 0$, $v_X^H > v_X^L > 0$, and $\Lambda_A > 0$. Then:*

1. *The hub-alone welfare gain is increasing in the share of high-absorptive participants:*

$$\frac{\partial \Delta_H(\sigma)}{\partial \sigma} = \bar{v}_L'(\sigma) P_A \Delta m_A \Lambda_A > 0. \tag{84}$$

2. *The optimal pipeline policy is increasing in σ:*

$$\frac{\partial b^*(\sigma)}{\partial \sigma} = \frac{\beta}{\kappa}\left[\xi \bar{v}_L'(\sigma) P_A m_A^1 + \bar{v}_X'(\sigma) Q_{AB}(\eta_0 + \eta_1)\right] > 0. \tag{85}$$

3. *The renewal surplus is increasing in σ:*

$$\frac{\partial R_1(\sigma)}{\partial \sigma} = \frac{\beta^2}{\kappa} A_1(\sigma)\left[\xi \bar{v}_L'(\sigma) P_A m_A^1 + \bar{v}_X'(\sigma) Q_{AB}(\eta_0 + \eta_1)\right] > 0. \tag{86}$$

Proposition 10 implies that the effective thickness of a regional ecosystem is not determined by headcount alone. Composition matters. A region with a relatively small number of participants may still support a valuable public hub if enough of those participants have high absorptive capacity or play boundary-spanning roles. Conversely, a larger ecosystem with weak absorptive capacity may fail to justify a hub despite its greater scale.

7.3 Implications of the Extensions

Taken together, the two extensions refine the baseline theory in complementary ways.

Partial internalization weakens the planner–decentralized wedge and narrows the corresponding support-threshold gap, thereby addressing the concern that the baseline benchmark is too stark. Higher absorptive capacity shifts the policy comparison in favor of both hubs and renewal by raising the value created from local and external interactions. The first extension is therefore a robustness check against the main critique of the benchmark, while the second is a regional-composition extension that helps interpret heterogeneous policy effects across places.

8 Empirical Predictions and Regional Policy Interpretation

The model is intended as a mechanism-based regional science framework rather than as a literal description of any single hub program. Its empirical value lies in clarifying which observable margins correspond to the theoretical objects introduced above, and in identifying predictions that distinguish *hub-alone coordination*, *pipeline-alone renewal*, and the *bundle* that combines the two. The relevant empirical baseline is therefore richer than a simple hub-versus-no-hub comparison.

A convenient starting point is the decomposition from Sections 5 and 6. The decentralized support criterion for adding a hub is

$$\Delta_P^* = S_A - F + \left(R_1^P - R_0^P\right),$$

while the planner's criterion is

$$\Delta_W^* = \Delta_P^* + D_A + M_A.$$

Here, S_A is the immediate local coordination gain, D_A is the dynamic local correction, and M_A is the additional social value of renewal that decentralized actors do not fully internalize. Likewise, Section 6 shows that

$$W^*(0) - W(0,0) = R_0 > 0$$

for pipeline policy alone, while

$$W^*(1) - W^*(0) = \Delta_H + \left(R_1 - R_0\right)$$

determines whether the full bundle is preferred to pipeline alone. These decompositions immediately suggest what should be measured empirically.

8.1 Mapping Theory to Observable Margins

Table 1 summarizes a parsimonious mapping from the model's theoretical objects to observable empirical margins.

The purpose of Table 1 is not to impose a single empirical design. Rather, it highlights that the model's core variables are observable at least in approximate form. In particular, a credible evaluation should include not only gross activity measures, but also indicators of novelty, repetition, and outside connectivity.

Some proxies are more tightly linked to particular parameters than others. First-time meetings, event-to-project conversion, and the share of new ties speak most directly to the hub-induced

Table 1: Illustrative Mapping from Theoretical Objects to Empirical Proxies

Theoretical object	Empirical interpretation	Illustrative proxies
Local coordination gain $m_A^1 - m_A^0$	Immediate effect of the hub on within-region matching	Interaction volume, first-time meetings, new project starts, event-to-project conversion
Dynamic redundancy δ	Speed at which repeated local interaction exhausts novelty	Repeated collaborator ratio, new ties share, decline in participant turnover, concentration of repeated ties
Renewal channel (ξ, η_0, η_1, b)	Extent to which outside brokerage renews the regional opportunity set and creates direct outside links	Outside-region collaborator share, invited outside participants, interregional project share, outside mentors, visiting researchers
Incremental bundle margin $R_1 - R_0$	Extra value of adding a hub to a renewal policy	Difference in medium-run novelty, outside collaboration, and repeated-tie dynamics between bundled and pipeline-only settings
Absorptive capacity / anchor presence	Capacity to convert contact and renewal into productive collaboration	Anchor participation share, university presence, lead-firm involvement, boundary-spanning intermediaries

matching increment $m_A^1 - m_A^0$. Repeated collaborator ratio and repeated-tie concentration are the closest empirical analogues of the redundancy parameter δ. Outside-link conversion and the persistence of new local combinations after outside contact are the most relevant proxies for the renewal channel, with the former most closely related to η_1 and the latter to ξ.

8.2 An Institutional Anchor

One reason to take the bundle logic seriously is that current policy practice already contains programs that combine a local support node with organized cross-regional access. A useful illustration is the European Digital Innovation Hubs (EDIH) network. The European Commission describes EDIHs as one-stop shops with a strong regional presence, but also emphasizes that each hub belongs to a wider pan-European network through which firms can access specialized expertise and services located outside their home region (European Commission, 2025). The institutional architecture is therefore explicitly two-layered: a regionally embedded access point plus network-based referral beyond the home region. In institutional terms, this is close to the distinction developed in the model: a local node reduces coordination frictions inside a region, while the wider network expands the region's effective outside pool.

Of course, EDIHs are not literal copies of the present model. They are digitally oriented service

infrastructures, not generic co-creation hubs, and their objectives are broader than the stylized welfare problem analyzed here. Nor do they by themselves validate the model's welfare ranking. But as an institutional fact they are useful because they show that policy practice already treats *local coordination* and *cross-regional access* as complementary rather than interchangeable. That is exactly the policy architecture the model is designed to clarify.

8.3 Three Testable Predictions

Prediction 1. Pipeline-only regions should show more outside connectivity than no-policy regions, but bundled regions should show more durable novelty than pipeline-only regions.

Section 6 shows that pipeline policy alone has positive value, summarized by $R_0 > 0$. This implies that even without a hub, a region using active brokerage should display more outside-region collaboration than an otherwise similar no-policy region.

However, the bundle is preferred to pipeline alone only when

$$\Delta_H + \left(R_1 - R_0\right) \geq 0.$$

This generates the paper's most identifying empirical comparison. Relative to pipeline-only regions, bundled regions should exhibit not just more outside collaboration, but also a more durable novelty profile: lower growth in repeated-tie concentration, higher persistence of new combinations, and stronger conversion of outside links into productive local collaboration.

Prediction 2. Hub creation raises visible local activity, but it need not preserve novelty over time.

The model implies an immediate coordination gain

$$B_1(1) - B_1(0) = v_L P_A\left(m_A^1 - m_A^0\right) > 0.$$

Thus, after hub establishment, one should expect more meetings, more matches, and more observed local project activity. However, the dynamic correction

$$D_A = \beta v_L P_A\left(m_A^1 - m_A^0\right)\left[1 - \delta\left(m_A^1 + m_A^0\right)\right]$$

need not be positive. When local redundancy is strong, early activity gains can coexist with a later rise in repeated collaborator ratios and a decline in the share of genuinely new ties.

The empirical implication is straightforward: evaluations based only on attendance, occupancy, or total interactions capture the visible coordination term but not whether novelty persists.

Prediction 3. The bundle advantage should be larger where outside renewal is easier and where absorptive capacity is higher.

The term $R_1 - R_0$ is increasing when outside renewal becomes more valuable. In the baseline model, this occurs when the outside pool is larger, brokerage is more effective, or renewal contributes more strongly to future local novelty. In the extension, the same logic is strengthened by higher absorptive capacity.

The empirical implication is that the same hub policy should look more favorable in regions with richer outside partner pools, stronger interregional brokerage institutions, or stronger anchor actors. Conversely, in places where outside renewal is weak, pipeline policy may still be worthwhile, but the incremental case for adding a hub should be weaker.

These three predictions are not equally diagnostic. Prediction 1 is the paper's most identifying implication because it distinguishes *pipeline alone* from the *bundle* and therefore matches the model's central policy comparison. Prediction 2 is easier to confirm empirically, but it is also less discriminating because many hub programs generate visible local activity. Prediction 3 is best interpreted as a moderator prediction: it identifies the regional conditions under which the bundled strategy should outperform pipeline policy alone by more.

8.4 Illustrative Parameter Regions

Figure 1 visualizes the model's main comparative policy implication. The dashed line is the hub-alone threshold F_H^* from Section 4. The solid curve is the bundled threshold

$$F_{H+b}^{\dagger}(N_B) = F_H^*(N_A) + \big(R_1(N_B) - R_0(N_B)\big)$$

from Section 6. For fixed costs between these two loci, a hub alone is not justified, but a hub embedded in a renewal strategy is preferred to pipeline policy alone.

Figure 1 should be read only as an illustration of the maintained functional form. It is not a calibration, an estimate, or a policy frontier recovered from data. Its role is simply to visualize how the conditional inequality in (57) partitions the (N_B, F) space under the benchmark assumptions.

8.5 Implications for Empirical Design

Taken together, the theoretical decomposition, the proxy mapping, and Figure 1 imply that a theory-consistent empirical evaluation of hub policy should be dynamic, network-aware, and explicitly regional.

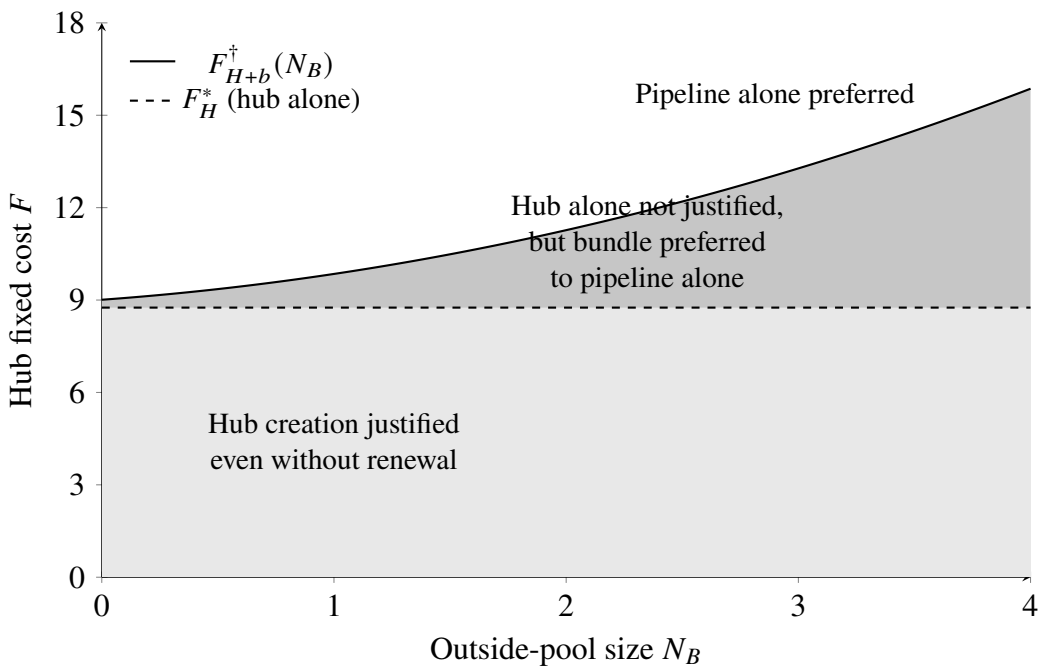

Figure 1: Illustrative policy regions in (N_B, F) space. The dashed line is the hub-alone threshold F^*_H. The solid curve is the bundled threshold $F^{\dagger}_{H+b}(N_B) = F^*_H + \big(R_1(N_B) - R_0(N_B)\big)$. Between the two loci, hub creation is justified only when embedded in a renewal strategy. Parameter values are illustrative: $N_A = 6$, $v_L = v_X = 1$, $m^0_A = 0.15$, $m^1_A = 0.55$, $\beta = 0.9$, $\delta = 0.70$, $\xi = 0.10$, $\eta_0 = 0.05$, $\eta_1 = 0.10$, and $\kappa = 1$.

First, evaluation should be *dynamic*. Early activity measures capture the immediate coordination gain, but do not show whether novelty persists. At a minimum, researchers should compare initial interaction gains with the subsequent evolution of new-ties share, repeated collaborator ratio, and participant turnover.

Second, evaluation should be *network-aware*. The model is fundamentally about the difference between creating interactions and creating *nonrepetitive* interactions. Measures such as repeated collaborator ratio, outside-region collaborator share, and the novelty content of joint applications or joint projects are therefore better aligned with the theory than aggregate attendance alone.

Third, evaluation should be *explicitly regional*. The outside pool is not a residual background feature. It is a policy-relevant object. Data on external participants, outside mentors, nonlocal co-applicants, or interregional partner flows are central outcome variables for judging whether a hub is functioning as part of a broader renewal strategy.

A minimal empirical design therefore requires more than a treated-versus-untreated comparison. The most informative setting is one in which researchers can distinguish among three baselines: no organized intervention, pipeline-oriented renewal without a hub, and the bundled strategy that combines local coordination with outside brokerage. In such a design, Prediction 1 is the most discriminating test. If the model is right, the strongest evidence for the bundled logic is not merely that hub regions become busier, but that bundled regions outperform pipeline-only regions on outside-link conversion, repeated-tie dynamics, and the persistence of genuinely new combinations.

In practice, the cleanest empirical settings would exploit staggered introductions of hub programs and interregional brokerage schemes, or program-level administrative data that distinguish local-only support from collaborative networking support. Panel event-study or matched-cohort designs could then trace not only activity effects, but also changes in outside-link conversion, repeated-tie concentration, and the persistence of new combinations across the three policy baselines. A minimal reduced-form specification would interact a bundle indicator with post-treatment time and test whether the coefficient on the bundle, relative to pipeline-only support, is positive for outside-link conversion and negative for repeated-tie concentration. This emphasis is consistent with the collaborative-policy literature: policy-induced R&D networks do not automatically generate stronger inter-regional knowledge diffusion (Bednarz and Broekel, 2019), and collaborative subsidies appear most relevant when they shift regions toward new technological combinations rather than merely increasing subsidized activity (Mewes and Broekel, 2020).

Overall, the model points toward a more demanding empirical agenda than a conventional utilization-based evaluation. The critical question is not simply whether the hub generated more activity, but whether it generated activity that remained novel, whether pipeline policy created

outside renewal even without a hub, and whether the bundle outperformed that pipeline-alone baseline.

9 Conclusion

This paper has developed a stylized two-region framework for evaluating public co-creation hubs in thin regional ecosystems. The model separates two policy margins that are often conflated in practice: *local coordination* inside the target region and *cross-regional renewal* through outside linkages. Its principal distortion is a dynamic coordination externality. Decentralized actors can value current coordination and directly contracted outside links, but they do not fully internalize how current local interaction changes the future novelty available to the regional system.

Three results follow. First, decentralized support and planner evaluation need not coincide. The same hub can be under-supported when renewal gains are neglected or over-supported when visible local activity masks future redundancy losses. Second, a hub is socially efficient as a stand-alone intervention only under limited conditions: its coordination gain must be large enough to offset fixed cost and the dynamic erosion of novelty generated by repeated local interaction. Third, pipeline policy has positive stand-alone value, so the relevant policy comparison is often not hub versus no hub, but pipeline alone versus a bundle that adds a hub to a renewal strategy. The bundle is preferred only under specific parameter conditions.

These conclusions imply a more demanding regional policy evaluation. A hub should not be judged only by attendance, occupancy, meetings, or other short-run utilization measures. A theory-consistent evaluation should also ask whether new ties continue to form, whether repeated collaborator ratios rise over time, whether outside-region collaboration expands under pipeline policy, and whether a bundled strategy outperforms a pipeline-only baseline. In regional science terms, the key issue is not merely whether a local venue is active, but whether the regional system combines internal coordination with enough external renewal to remain dynamically productive in the model's minimal two-period sense.

For that reason, the most informative empirical setting is not a simple treated-versus-untreated comparison, but a three-way comparison among no organized intervention, pipeline-oriented renewal without a hub, and the bundled strategy. Only that design can show whether a hub adds value beyond an already active renewal policy.

The framework is intentionally parsimonious, and that parsimony defines its limits. Matching rates, outside-link productivity, and the outside pool are all reduced-form objects. Network formation is not endogenous, and neither migration nor location choice is modeled. The analysis

is therefore best interpreted as a conditional policy theory rather than as a full structural account of regional evolution. Its purpose is to isolate one principal welfare wedge and show how that wedge changes the evaluation of hub policy in thin regional ecosystems. The partial-internalization extension also makes clear that the benchmark is not all-or-nothing: as decentralized actors internalize more of the continuation value, the support-threshold gap shrinks and vanishes in the full-internalization limit.

Within those limits, the paper offers a conditional implication for the design of place-based policy. In thin regional ecosystems, public hubs are most plausibly justified not as stand-alone venues that generate visible activity, but as elements of a broader regional strategy that combines local coordination with cross-regional renewal. Whether such a bundle is desirable remains conditional, but the relevant benchmark for judging hub policy changes once renewal is treated as a policy margin in its own right.

Data availability

This study develops a theoretical model and does not analyze newly collected empirical data. No dataset accompanies the manuscript.

Statements and Declarations

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A Proofs for Section 4

This appendix collects the algebra omitted from Section 4.

A.1 Preliminaries

Recall from Section 3 that

$$P_A = \frac{N_A(N_A - 1)}{2}, \qquad Q_{AB} = N_A N_B, \qquad m_A(h) = m_A^0 + \left(m_A^1 - m_A^0\right)h,$$

with

$$0 \le m_A^0 < m_A^1 \le 1.$$

In the benchmark case without pipeline policy ($b = 0$), welfare is

$$W(0,0) = v_L P_A m_A^0 + \beta\left\{v_L P_A m_A^0\left(1 - \delta m_A^0\right) + v_X Q_{AB} q_0\right\}, \tag{87}$$

and

$$W(1,0) = v_L P_A m_A^1 + \beta\left\{v_L P_A m_A^1\left(1 - \delta m_A^1\right) + v_X Q_{AB} q_0\right\} - F. \tag{88}$$

Hence the net welfare gain from a hub alone is

$$\begin{aligned}\Delta_H &:= W(1,0) - W(0,0)\\ &= v_L P_A \Big[m_A^1 - m_A^0 + \beta\big(m_A^1(1-\delta m_A^1) - m_A^0(1-\delta m_A^0)\big)\Big] - F. \end{aligned} \tag{89}$$

The cross-regional baseline term cancels exactly:

$$\beta v_X Q_{AB} q_0 - \beta v_X Q_{AB} q_0 = 0.$$

Thus the hub-alone comparison depends only on local primitives.

Now,

$$\begin{aligned} m_A^1(1-\delta m_A^1) - m_A^0(1-\delta m_A^0) &= (m_A^1 - m_A^0) - \delta\big((m_A^1)^2 - (m_A^0)^2\big)\\ &= (m_A^1 - m_A^0)\big[1 - \delta(m_A^1 + m_A^0)\big]. \end{aligned} \tag{90}$$

Substituting (90) into (89) yields

$$\Delta_H = v_L P_A (m_A^1 - m_A^0)\Big[(1+\beta) - \beta\delta(m_A^1 + m_A^0)\Big] - F. \tag{91}$$

Define

$$\Lambda_A := (1+\beta) - \beta\delta(m_A^1 + m_A^0), \tag{92}$$

so that

$$\Delta_H = v_L P_A (m_A^1 - m_A^0)\Lambda_A - F. \tag{93}$$

A.2 Proof of Proposition 1

We prove each part in turn.

Part 1: the case $\Lambda_A \le 0$. Since $v_L > 0$, $P_A > 0$ for all $N_A \ge 2$, and $m_A^1 - m_A^0 > 0$, the product

$$v_L P_A (m_A^1 - m_A^0)\Lambda_A$$

is weakly negative whenever $\Lambda_A \le 0$. Therefore, from (93),

$$\Delta_H = v_L P_A (m_A^1 - m_A^0)\Lambda_A - F < 0$$

for every $F > 0$. Hence the hub is never socially efficient.

Part 2: the case $\Lambda_A > 0$. Under $\Lambda_A > 0$, the condition $\Delta_H \geq 0$ is equivalent to

$$v_L P_A (m_A^1 - m_A^0)\Lambda_A \geq F.$$

Since all terms except P_A are strictly positive, this can be rearranged as

$$P_A \geq \frac{F}{v_L (m_A^1 - m_A^0)\Lambda_A}. \tag{94}$$

This proves (21).

Using $P_A = N_A(N_A - 1)/2$, (94) becomes

$$\frac{N_A(N_A - 1)}{2} \geq \frac{F}{v_L (m_A^1 - m_A^0)\Lambda_A},$$

or equivalently

$$N_A^2 - N_A - \frac{2F}{v_L (m_A^1 - m_A^0)\Lambda_A} \geq 0.$$

The positive root of the corresponding quadratic equation is

$$\frac{1}{2}\left(1 + \sqrt{1 + \frac{8F}{v_L (m_A^1 - m_A^0)\Lambda_A}}\right).$$

Hence, treating N_A as continuous, the hub is efficient if and only if

$$N_A \geq N_H^* := \frac{1}{2}\left(1 + \sqrt{1 + \frac{8F}{v_L (m_A^1 - m_A^0)\Lambda_A}}\right), \tag{95}$$

which is exactly (22). For integer N_A, replace N_H^* by its ceiling. This completes the proof. □

A.3 Derivation of the Equivalent Threshold Characterizations

Starting from (93), the condition $\Delta_H \geq 0$ is equivalent to

$$F \leq v_L P_A (m_A^1 - m_A^0)\Lambda_A.$$

Thus the critical fixed cost is

$$F_H^*(N_A) := v_L P_A (m_A^1 - m_A^0)\Lambda_A, \tag{96}$$

which is (23).

Next, solve for the critical redundancy rate. Rewrite $\Delta_H \geq 0$ as

$$v_L P_A (m_A^1 - m_A^0)\Big[(1+\beta) - \beta\delta(m_A^1 + m_A^0)\Big] \geq F.$$

Dividing by the positive term $v_L P_A (m_A^1 - m_A^0)$ gives

$$(1+\beta) - \beta\delta(m_A^1 + m_A^0) \geq \frac{F}{v_L P_A (m_A^1 - m_A^0)}.$$

Hence

$$\beta\delta(m_A^1 + m_A^0) \leq (1+\beta) - \frac{F}{v_L P_A (m_A^1 - m_A^0)},$$

so

$$\begin{aligned}\delta &\leq \frac{1+\beta}{\beta(m_A^1 + m_A^0)} - \frac{F}{\beta v_L P_A (m_A^1 - m_A^0)(m_A^1 + m_A^0)} \\ &=: \delta_H^*(N_A), \end{aligned} \tag{97}$$

which is (24).

A.4 Proof of Proposition 2

Throughout this proof we use

$$\Delta_H = v_L P_A (m_A^1 - m_A^0)\Lambda_A - F, \qquad \Lambda_A = (1+\beta) - \beta\delta(m_A^1 + m_A^0).$$

Derivative with respect to N_A. Treating N_A as continuous,

$$P_A = \frac{N_A(N_A - 1)}{2} \quad \Rightarrow \quad \frac{\partial P_A}{\partial N_A} = \frac{2N_A - 1}{2}.$$

Hence

$$\begin{aligned}\frac{\partial \Delta_H}{\partial N_A} &= v_L \frac{\partial P_A}{\partial N_A}(m_A^1 - m_A^0)\Lambda_A \\ &= \frac{v_L}{2}(2N_A - 1)(m_A^1 - m_A^0)\Lambda_A, \end{aligned} \tag{98}$$

which is (25). If $\Lambda_A > 0$, then $\partial\Delta_H/\partial N_A > 0$ because $v_L > 0$, $2N_A - 1 > 0$, and $m_A^1 - m_A^0 > 0$.

For integer N_A, the same monotonicity follows from forward differences:

$$\begin{aligned}\Delta_H(N_A+1)-\Delta_H(N_A) &= v_L\big(P_A(N_A+1)-P_A(N_A)\big)(m_A^1-m_A^0)\Lambda_A \\ &= v_L N_A(m_A^1-m_A^0)\Lambda_A > 0 \quad \text{if } \Lambda_A > 0. \end{aligned} \tag{99}$$

Derivative with respect to F. From (93),

$$\frac{\partial \Delta_H}{\partial F} = -1 < 0, \tag{100}$$

which is (26).

Derivative with respect to δ. Differentiating $\Lambda_A = (1+\beta) - \beta\delta(m_A^1 + m_A^0)$ gives

$$\frac{\partial \Lambda_A}{\partial \delta} = -\beta(m_A^1 + m_A^0).$$

Therefore

$$\begin{aligned}\frac{\partial \Delta_H}{\partial \delta} &= v_L P_A(m_A^1 - m_A^0)\frac{\partial \Lambda_A}{\partial \delta} \\ &= -\beta v_L P_A(m_A^1 - m_A^0)(m_A^1 + m_A^0) < 0, \end{aligned} \tag{101}$$

which is (27).

Derivative with respect to m_A^1. Starting from

$$\Delta_H = v_L P_A(m_A^1 - m_A^0)\Big[(1+\beta) - \beta\delta(m_A^1 + m_A^0)\Big] - F,$$

differentiate with respect to m_A^1:

$$\begin{aligned}\frac{\partial \Delta_H}{\partial m_A^1} &= v_L P_A\left(\Big[(1+\beta) - \beta\delta(m_A^1 + m_A^0)\Big] + (m_A^1 - m_A^0)(-\beta\delta)\right) \\ &= v_L P_A\Big[(1+\beta) - 2\beta\delta m_A^1\Big]. \end{aligned} \tag{102}$$

This is exactly (28).

To verify the sign, note that the regularity condition (6) implies

$$1 - \delta m_A^1 > 0 \quad \Longrightarrow \quad \delta m_A^1 < 1.$$

Hence

$$2\beta\delta m_A^1 < 2\beta.$$

Since $\beta \in (0, 1]$, we have $1 + \beta \geq 2\beta$, with equality only when $\beta = 1$. Therefore,

$$(1 + \beta) - 2\beta\delta m_A^1 > (1 + \beta) - 2\beta = 1 - \beta \geq 0,$$

where the strict inequality follows from $\delta m_A^1 < 1$. Thus

$$\frac{\partial \Delta_H}{\partial m_A^1} > 0. \tag{103}$$

Derivative with respect to m_A^0. Similarly,

$$\begin{aligned}\frac{\partial \Delta_H}{\partial m_A^0} &= v_L P_A \left(-\left[(1 + \beta) - \beta\delta(m_A^1 + m_A^0)\right] + (m_A^1 - m_A^0)(-\beta\delta)\right) \\ &= -v_L P_A \left[(1 + \beta) - 2\beta\delta m_A^0\right]. \end{aligned} \tag{104}$$

This is exactly (29).

Because $m_A^0 < m_A^1$ and $\delta m_A^1 < 1$, we also have $\delta m_A^0 < 1$. Repeating the same sign argument as above gives

$$(1 + \beta) - 2\beta\delta m_A^0 > 0,$$

so

$$\frac{\partial \Delta_H}{\partial m_A^0} < 0. \tag{105}$$

This proves Proposition 2. □

B Proofs for Section 5

This appendix provides the algebra omitted from Section 5.

B.1 Proof of Proposition 3

Fix $h \in \{0, 1\}$. From (15), the decentralized benchmark solves

$$\max_{b \geq 0} \left\{ \bar{\Pi}(h) + \beta A_h^P b - \frac{\kappa}{2} b^2 \right\}.$$

Differentiating with respect to b gives

$$\frac{\partial \Pi(h,b)}{\partial b} = \beta A_h^P - \kappa b.$$

Since

$$\frac{\partial^2 \Pi(h,b)}{\partial b^2} = -\kappa < 0,$$

the problem is strictly concave, and the first-order condition is sufficient. Setting the derivative equal to zero yields

$$b_P^*(h) = \frac{\beta A_h^P}{\kappa} = \frac{\beta v_X Q_{AB}(\eta_0 + \eta_1 h)}{\kappa},$$

which is (30).

Likewise, from (16), the planner solves

$$\max_{b \geq 0} \left\{ \bar{W}(h) + \beta A_h^W b - \frac{\kappa}{2} b^2 \right\}.$$

Differentiating,

$$\frac{\partial W(h,b)}{\partial b} = \beta A_h^W - \kappa b, \qquad \frac{\partial^2 W(h,b)}{\partial b^2} = -\kappa < 0,$$

so the unique optimum is

$$b_W^*(h) = \frac{\beta A_h^W}{\kappa} = \frac{\beta \Big[\xi v_L P_A m_A(h) + v_X Q_{AB}(\eta_0 + \eta_1 h) \Big]}{\kappa},$$

which is (31).

Subtracting the two solutions gives

$$b_W^*(h) - b_P^*(h) = \frac{\beta \big(A_h^W - A_h^P \big)}{\kappa} = \frac{\beta \xi v_L P_A m_A(h)}{\kappa} > 0,$$

which proves (32). Using $m_A(1) - m_A(0) = m_A^1 - m_A^0 > 0$, we also obtain

$$\big[b_W^*(1) - b_P^*(1) \big] - \big[b_W^*(0) - b_P^*(0) \big] = \frac{\beta \xi v_L P_A \big(m_A^1 - m_A^0 \big)}{\kappa} > 0,$$

which is (33). This proves Proposition 3. □

B.2 Derivation of the Decentralized–Social Wedge

Substituting the optimal pipeline choices into (15) and (16) gives

$$\Pi^*(h) = \bar{\Pi}(h) + \frac{\beta^2 (A_h^P)^2}{2\kappa} = \bar{\Pi}(h) + R_h^P,$$

and

$$W^*(h) = \bar{W}(h) + \frac{\beta^2 (A_h^W)^2}{2\kappa} = \bar{W}(h) + R_h^W.$$

Using (17) and (15),

$$\bar{W}(1) - \bar{W}(0) = v_L P_A (m_A^1 - m_A^0) + \beta v_L P_A \left[m_A^1 (1 - \delta m_A^1) - m_A^0 (1 - \delta m_A^0) \right] - F.$$

Factoring the second term,

$$\bar{W}(1) - \bar{W}(0) = S_A + D_A - F,$$

where S_A and D_A are defined in (38) and (39). Also,

$$\bar{\Pi}(1) - \bar{\Pi}(0) = S_A - F.$$

Therefore

$$\Delta_P^* = \Pi^*(1) - \Pi^*(0) = S_A - F + (R_1^P - R_0^P),$$

which is (41), and

$$\Delta_W^* = W^*(1) - W^*(0) = S_A + D_A - F + (R_1^W - R_0^W).$$

Subtracting the two expressions gives

$$\Delta_W^* - \Delta_P^* = D_A + \left[(R_1^W - R_0^W) - (R_1^P - R_0^P) \right] = D_A + M_A,$$

which is (42).

To verify $M_A > 0$, note that

$$A_h^W - A_h^P = \xi v_L P_A m_A(h) > 0 \qquad \text{for } h \in \{0, 1\},$$

and that both $m_A(h)$ and A_h^P are weakly larger when $h = 1$ than when $h = 0$. Hence

$$(A_1^W)^2 - (A_1^P)^2 > (A_0^W)^2 - (A_0^P)^2,$$

which implies

$$M_A = \frac{\beta^2}{2\kappa}\left((A_1^W)^2 - (A_1^P)^2 - (A_0^W)^2 + (A_0^P)^2\right) > 0.$$

B.3 Proof of Proposition 4

By definition,

$$\Delta_P^* > 0 \quad \Longleftrightarrow \quad F < F_P^*,$$

where

$$F_P^* = S_A + (R_1^P - R_0^P).$$

Likewise,

$$\Delta_W^* > 0 \quad \Longleftrightarrow \quad F < F_W^*,$$

where

$$F_W^* = F_P^* + D_A + M_A.$$

Part 1. If $D_A + M_A > 0$, then $F_W^* > F_P^*$. Therefore the interval

$$F_P^* < F < F_W^*$$

is nonempty. On this interval,

$$\Delta_P^* < 0 \qquad \text{and} \qquad \Delta_W^* > 0.$$

Hence decentralized actors reject the hub while the planner supports it. This proves (46).

Part 2. If $D_A + M_A < 0$, then $F_W^* < F_P^*$. Therefore the interval

$$F_W^* < F < F_P^*$$

is nonempty. On this interval,

$$\Delta_W^* < 0 \qquad \text{and} \qquad \Delta_P^* > 0.$$

Hence decentralized actors support the hub while the planner rejects it. This proves (48).

This completes the proof of Proposition 4. □

C Proofs for Section 6

This appendix provides the proofs omitted from Section 6.

C.1 Proof of Proposition 5

Fix $h \in \{0, 1\}$. From (16), the planner solves

$$\max_{b \geq 0} \left\{ \bar{W}(h) + \beta A_h^W b - \frac{\kappa}{2} b^2 \right\}.$$

Differentiating with respect to b,

$$\frac{\partial W(h, b)}{\partial b} = \beta A_h^W - \kappa b.$$

The second derivative is

$$\frac{\partial^2 W(h, b)}{\partial b^2} = -\kappa < 0,$$

so the problem is strictly concave and the first-order condition is sufficient. Setting the derivative equal to zero gives

$$b^*(h) = \frac{\beta A_h^W}{\kappa},$$

which is (49).

Substituting this optimizer back into the objective yields

$$\begin{aligned} W^*(h) &= \bar{W}(h) + \beta A_h^W \left(\frac{\beta A_h^W}{\kappa} \right) - \frac{\kappa}{2} \left(\frac{\beta A_h^W}{\kappa} \right)^2 \\ &= \bar{W}(h) + \frac{\beta^2 (A_h^W)^2}{\kappa} - \frac{\beta^2 (A_h^W)^2}{2\kappa} \\ &= \bar{W}(h) + \frac{\beta^2 (A_h^W)^2}{2\kappa} = \bar{W}(h) + R_h, \end{aligned}$$

which proves (50). This establishes Proposition 5. □

C.2 Proof of Proposition 6

From (53),

$$W^*(0) - W(0, 0) = R_0.$$

If $A_0^W > 0$, then by definition

$$R_0 = \frac{\beta^2 (A_0^W)^2}{2\kappa} > 0,$$

which proves (56).

Next, from (55) and (53),

$$W^*(1) - W^*(0) = \Delta_H + R_1 - R_0.$$

Therefore the bundle is preferred to pipeline alone if and only if

$$\Delta_H + (R_1 - R_0) \geq 0,$$

which is (57).

From Section 4,

$$\Delta_H = F_H^*(N_A) - F.$$

Substituting into the previous condition gives

$$F_H^*(N_A) - F + R_1 - R_0 \geq 0,$$

or equivalently

$$F \leq F_H^*(N_A) + (R_1 - R_0) = F_{H+b}^\dagger(N_A, N_B),$$

which is (58).

Finally, since (51) states that $R_1 > R_0$, we obtain

$$F_{H+b}^\dagger(N_A, N_B) - F_H^*(N_A) = R_1 - R_0 > 0,$$

which is (59). This completes the proof of Proposition 6. □

C.3 Proof of Proposition 7

A hub alone is inefficient if and only if

$$\Delta_H < 0 \quad \Longleftrightarrow \quad F > F_H^*(N_A).$$

The bundle is preferred to pipeline alone if and only if

$$\Delta_H + (R_1 - R_0) \geq 0 \quad \Longleftrightarrow \quad F \leq F_H^*(N_A) + (R_1 - R_0).$$

Combining the two conditions gives

$$F_H^*(N_A) < F \leq F_H^*(N_A) + (R_1 - R_0),$$

which is exactly (60). Because $R_1 > R_0$, the interval has positive length. This proves Proposition 7. □

D Proofs for Section 7

This appendix provides the algebra for the extensions in Section 7.

D.1 Preliminaries

Recall the baseline objects

$$P_A = \frac{N_A(N_A - 1)}{2}, \qquad Q_{AB} = N_A N_B,$$

$$\Delta m_A := m_A^1 - m_A^0 > 0, \qquad \Lambda_A := (1 + \beta) - \beta\delta(m_A^1 + m_A^0).$$

D.2 Proof of Proposition 8

Fix $h \in \{0, 1\}$ and $\alpha \in [0, 1]$. From (63), the decentralized benchmark solves

$$\max_{b \geq 0} \left\{ \bar{\Pi}_\alpha(h) + \beta A_{h,\alpha}^P b - \frac{\kappa}{2} b^2 \right\}.$$

Differentiating with respect to b gives

$$\frac{\partial \Pi_\alpha(h, b)}{\partial b} = \beta A_{h,\alpha}^P - \kappa b.$$

Because

$$\frac{\partial^2 \Pi_\alpha(h, b)}{\partial b^2} = -\kappa < 0,$$

the problem is strictly concave and the first-order condition is sufficient. Setting the derivative equal to zero yields

$$b^*_{P,\alpha}(h) = \frac{\beta A^P_{h,\alpha}}{\kappa},$$

which is (65).

Next, compare this with the planner's choice from (31):

$$\begin{aligned} b^*_W(h) - b^*_{P,\alpha}(h) &= \frac{\beta}{\kappa}\left[\xi v_L P_A m_A(h) + v_X Q_{AB}(\eta_0 + \eta_1 h)\right] \\ &\quad - \frac{\beta}{\kappa}\left[\alpha\xi v_L P_A m_A(h) + v_X Q_{AB}(\eta_0 + \eta_1 h)\right] \\ &= \frac{\beta(1-\alpha)\xi v_L P_A m_A(h)}{\kappa}, \end{aligned} \tag{106}$$

which proves (66). The inequality is strict for every $\alpha < 1$ because β, ξ, v_L, P_A, and $m_A(h)$ are strictly positive.

Finally, differentiating (65) with respect to α gives

$$\frac{\partial b^*_{P,\alpha}(h)}{\partial \alpha} = \frac{\beta\xi v_L P_A m_A(h)}{\kappa} > 0,$$

which proves (67). This completes the proof of Proposition 8. □

D.3 Proof of Proposition 9

Recall from Section 5 that

$$S_A = v_L P_A(m^1_A - m^0_A), \qquad D_A = \beta v_L P_A(m^1_A - m^0_A)\left[1 - \delta(m^1_A + m^0_A)\right].$$

Under partial internalization, the decentralized support threshold is

$$F^\dagger_{P,\alpha} = S_A + \alpha D_A + \left(R^P_{1,\alpha} - R^P_{0,\alpha}\right),$$

while the planner's threshold remains

$$F^\dagger_{H+b} = S_A + D_A + \left(R_1 - R_0\right).$$

Subtracting the two thresholds gives

$$\begin{aligned}\Omega_\alpha &= F^\dagger_{H+b} - F^\dagger_{P,\alpha} \\ &= (1-\alpha)D_A\Big[(R_1 - R_0) - \big(R^P_{1,\alpha} - R^P_{0,\alpha}\big)\Big].\end{aligned} \tag{107}$$

Write

$$a_h := \xi v_L P_A m_h, \qquad c_h := v_X Q_{AB}(\eta_0 + \eta_1 h), \qquad m_h = m^0_A + (m^1_A - m^0_A)h.$$

Then

$$R_h = \frac{\beta^2(a_h + c_h)^2}{2\kappa}, \qquad R^P_{h,\alpha} = \frac{\beta^2(\alpha a_h + c_h)^2}{2\kappa}.$$

Hence

$$\begin{aligned}R_h - R^P_{h,\alpha} &= \frac{\beta^2}{2\kappa}\Big[(a_h + c_h)^2 - (\alpha a_h + c_h)^2\Big] \\ &= \frac{\beta^2}{2\kappa}(1-\alpha)a_h\big[(1+\alpha)a_h + 2c_h\big].\end{aligned} \tag{108}$$

Applying (108) at $h = 1$ and $h = 0$ yields

$$(R_1 - R_0) - \big(R^P_{1,\alpha} - R^P_{0,\alpha}\big) = \frac{\beta^2}{2\kappa}(1-\alpha)\left[(1+\alpha)(a_1^2 - a_0^2) + 2(a_1c_1 - a_0c_0)\right]. \tag{109}$$

Now

$$a_1^2 - a_0^2 = \xi^2 v_L^2 P_A^2\big((m^1_A)^2 - (m^0_A)^2\big) = X_A,$$

and

$$2(a_1c_1 - a_0c_0) = 2\xi v_L P_A v_X Q_{AB}\Big[\eta_0(m^1_A - m^0_A) + \eta_1 m^1_A\Big] = Y_A.$$

Substituting these identities into (107) and (109) gives

$$\Omega_\alpha = (1-\alpha)\left[D_A + \frac{\beta^2}{2\kappa}\Big((1+\alpha)X_A + Y_A\Big)\right],$$

which is (73).

If $\Omega_\alpha > 0$, then

$$F^\dagger_{P,\alpha} < F^\dagger_{H+b},$$

so every fixed cost in the interval (74) yields decentralized rejection and planner support. If $\Omega_\alpha < 0$, the inequality is reversed, giving the interval (75). This proves parts (1) and (2).

Part (3) follows immediately from (73): setting $\alpha = 1$ gives $\Omega_1 = 0$. Differentiating (73) with respect to α gives

$$\frac{\partial \Omega_\alpha}{\partial \alpha} = -D_A - \frac{\beta^2}{2\kappa}\Big(2\alpha X_A + Y_A\Big). \tag{110}$$

Because $X_A > 0$, $Y_A > 0$, and $\alpha \in [0, 1)$, the second term on the right-hand side is strictly negative. Therefore, if $D_A \geq 0$, then

$$\frac{\partial \Omega_\alpha}{\partial \alpha} < 0 \qquad \text{for all } \alpha \in [0, 1),$$

so the threshold gap is strictly decreasing in α. This completes the proof of Proposition 9. □

D.4 Proof of Proposition 10

In the heterogeneous-participant extension, the expected surplus per realized local interaction is

$$\bar{v}_L(\sigma) = v_L^L(1-\sigma)^2 + v_L^H\big[1-(1-\sigma)^2\big] = v_L^L + (2\sigma - \sigma^2)(v_L^H - v_L^L), \tag{111}$$

with

$$\bar{v}_L'(\sigma) = 2(1-\sigma)(v_L^H - v_L^L) > 0 \qquad \text{for } \sigma \in [0, 1). \tag{112}$$

Because $\bar{v}_L(\sigma)$ is a convex combination of v_L^L and v_L^H,

$$\bar{v}_L(\sigma) > 0 \qquad \text{for all } \sigma \in [0, 1]. \tag{113}$$

Likewise,

$$\bar{v}_X(\sigma) = v_X^L(1-\sigma) + v_X^H\sigma = v_X^L + \sigma(v_X^H - v_X^L), \tag{114}$$

with

$$\bar{v}_X'(\sigma) = v_X^H - v_X^L > 0 \tag{115}$$

and

$$\bar{v}_X(\sigma) > 0 \qquad \text{for all } \sigma \in [0, 1]. \tag{116}$$

Using (77), the hub-alone welfare gain is

$$\Delta_H(\sigma) = \bar{v}_L(\sigma) P_A \Delta m_A \Lambda_A - F.$$

Differentiating with respect to σ gives

$$\frac{\partial \Delta_H(\sigma)}{\partial \sigma} = \bar{v}_L'(\sigma) P_A \Delta m_A \Lambda_A. \tag{117}$$

If $\Lambda_A > 0$, then (112) implies

$$\frac{\partial \Delta_H(\sigma)}{\partial \sigma} > 0 \qquad \text{for } \sigma \in [0, 1),$$

which proves (84).

Next, from (82),

$$b^*(\sigma) = \frac{\beta A_1(\sigma)}{\kappa}, \qquad A_1(\sigma) = \xi \bar{v}_L(\sigma) P_A m_A^1 + \bar{v}_X(\sigma) Q_{AB}(\eta_0 + \eta_1).$$

Differentiating,

$$\frac{\partial b^*(\sigma)}{\partial \sigma} = \frac{\beta}{\kappa} \left[\xi \bar{v}_L'(\sigma) P_A m_A^1 + \bar{v}_X'(\sigma) Q_{AB}(\eta_0 + \eta_1) \right]. \tag{118}$$

Since all multiplicative terms are positive, (112) and (115) imply

$$\frac{\partial b^*(\sigma)}{\partial \sigma} > 0,$$

which proves (85).

For the renewal surplus,

$$R_1(\sigma) = \frac{\beta^2 A_1(\sigma)^2}{2\kappa}.$$

Differentiating,

$$\frac{\partial R_1(\sigma)}{\partial \sigma} = \frac{\beta^2}{\kappa} A_1(\sigma) \left[\xi \bar{v}_L'(\sigma) P_A m_A^1 + \bar{v}_X'(\sigma) Q_{AB}(\eta_0 + \eta_1) \right]. \tag{119}$$

By (113), (116), (112), and (115), every factor on the right-hand side is strictly positive. Hence

$$\frac{\partial R_1(\sigma)}{\partial \sigma} > 0,$$

which proves (86). This completes the proof of Proposition 10. □